

# Navid Bahadoran

(425)209-7391 | Tallahassee, FL | navid.bahadoran@gmail.com | Personal Website | LinkedIn | GitHub | US Citizen

## EDUCATION

**Florida State University (FSU)**, *PhD in Applied Mathematics, Financial Mathematics* (GPA: 3.91/4) **Expected May 2028**  
**University of Washington (UW)**, *MS in Applied Mathematics, Financial Mathematics* (GPA: 3.95/4) **Dec 2021**  
**Sharif University of Technology**, *BS in Electrical Engineering*

## TECHNICAL SKILLS

- Programming / Data: Python, SQL, R, Git, C++ (basic)
- Quantitative Methods: Statistical inference, time-series & panel econometrics, factor models, PCA, Monte Carlo simulation, covariance estimation, optimization, regularization, model evaluation
- Machine Learning: Regression, classification, boosting, cross-validation, feature engineering, model evaluation; NumPy, pandas, scikit-learn, PyTorch, XGBoost

## WORK EXPERIENCE

**State of Wisconsin Investment Board (SWIB)**: Madison, WI **2021 – 2023**

*Data Analyst, Risk Analytics and Systems*

- Built Python-based analytics pipelines for ex-ante portfolio risk measurement using factor-model exposures and Monte Carlo simulation.
- Designed enterprise liquidity data infrastructure using SQL and SQLAlchemy to integrate portfolio and risk data supporting liquidity and risk analytics.

**Massachusetts Pension Reserves Investment Management Board (Mass PRIM)**: Boston, MA **2021**

*Quantitative Research Intern*

- Conducted quantitative research on public companies using historical fundamentals and statistical classification models.
- Implemented Python-based workflows to identify candidate firms and evaluate predictive investment signals.

**T-Mobile: Bellevue, WA**

**2014 - 2020**

*Network Systems Engineer*

- Developed Python-based risk-management tooling and quantitative analyses on large operational datasets.

## SELECTED QUANTITATIVE RESEARCH

**Debiased Inference for Dynamic Panels with Low-Rank Coefficient Heterogeneity – FSU** **2026 - Present**

- Developing high-dimensional dynamic-panel models with heterogeneous coefficients, latent common factors, and cross-sectional dependence.
- Developing low-rank estimation, rank-selection, and bias-corrected inference methods; evaluating robustness through Monte Carlo experiments under cross-sectional dependence.
- Evaluating finite-sample bias, rank-selection stability, and inference under heteroskedasticity, predetermined covariates, and cross-sectional dependence.

**Leakage-Aware PCA for High-Dimensional Factor Models – FSU** **2026 - Present**

- Developing high-dimensional PCA and factor-model methods that distinguish population signal from same-sample target leakage and estimation error.
- Designing repeated out-of-fold and purged validation to assess model-selection risk, refit stability, and out-of-sample reconstruction in rolling macroeconomic data.
- Evaluating leakage-aware calibration through simulations and rolling U.S. macroeconomic data with purged tuning and test blocks

**High-Dimensional Portfolio Optimization – FSU** **2025**

- Developed PCA-based dimensionality-reduction and covariance-estimation methods for asset returns and evaluated out-of-sample portfolio allocation.
- Evaluated the sensitivity of portfolio risk and allocation to covariance-estimation error and dimensionality-reduction choices..

**Firm Bankruptcy Prediction – FSU** **2024**

- Built Python-based statistical and machine-learning models for corporate distress prediction using historical firm data.
- Compared SVM, random forest, k-nearest neighbors, and neural networks using cross-validation, AUC, and F1.

## AWARDS AND ACTIVITIES

Dean Award for Doctoral Excellence, FSU | Bettye Anne Graduate Fellowship, FSU | Tam Family Poster Competition Award, FSU | Academic Achievement and Peer Leadership Award – UW | Chess, Tennis, Pickleball, Swimming; Piano and Santoor